Near-Null Magnetic Fields in Lunar Plant Biology: The First 5 Days of Growth

1. LUNAR MAGNETIC REALITY

- Moon lacks global core dynamo
- Artemis South Pole: < 0.8 nT (Deep near-null magnetic void)
- Microclimatic field variation

Payload Field Architecture:

Btotal = Bcrust + Bext + Blander

Lander Avionics (50–500 μT)
Swamps Ambient Lunar HMF
⟶ In Situ Sensor Required

Planetary Environment:
Extreme Lunar HMF (<0.8 nT)

2. FIRST 5 DAYS OF GROWTH

- 0–72 h Heterotrophic Bottleneck
- Peroxisomal beta-oxidation
- Mitochondrial restart burst
- Plastid greening transition

Empirical Phenotypes in NNMF:

- Viability: -10 to -30% (# Plants)
- Velocity: Δt50 = +16 h Lag
- Dry Biomass: -28.5% Penalty
- Pigments: HEMA1, CHLH, POR & HY5 Repressed (-1.8 to -2.5x)

Developmental Penalty:
Viability, Speed, Pigments

3. RISK & SCIENCE FRAMEWORK

- Exploration Atmosphere (8.2 psia)
- Elevated CO2 (1200–3000 ppm)
- GCR Ionizing Radiation (1 Gy)
- Shared organellar bioenergetics

Rigorous Science & Flight Strategy:

- Ground Rigor: 3-chamber & bifilar sham controls + causal rescue
- Flight Payloads: Realistic 3-axis canopy magnetic vector logging
- Science: Phenotype disambiguation

Operational Translation:
Ground Rigor & Flight Logging